

END-TO-END INTELLIGENCE PLATFORM

# EduShield

## Academic Integrity Intelligence Platform

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Synthetic Data • Machine Learning • MySQL  
Streamlit • Power BI

End-to-End Data Science & Business Intelligence Project



# The Challenge

- 01 AI-generated content detection complexity
- 02 Multiple detector signals integration
- 03 Submission behavior patterns
- 04 Writing characteristics analysis
- 05 Academic context influence
- 06 Historical integrity events
- 07 Investigation records & human review



# What EduShield Does

01

**Data Engineering**

Relational pipelines

02

**Data Cleaning**

Trusted analytical layer

03

**AI-Generation Prediction**

Leakage-aware screening

04

**Detector Intelligence**

Signal comparison

05

**Behavioral Analytics**

Submission patterns

06

**Power BI Intelligence**

Executive insight

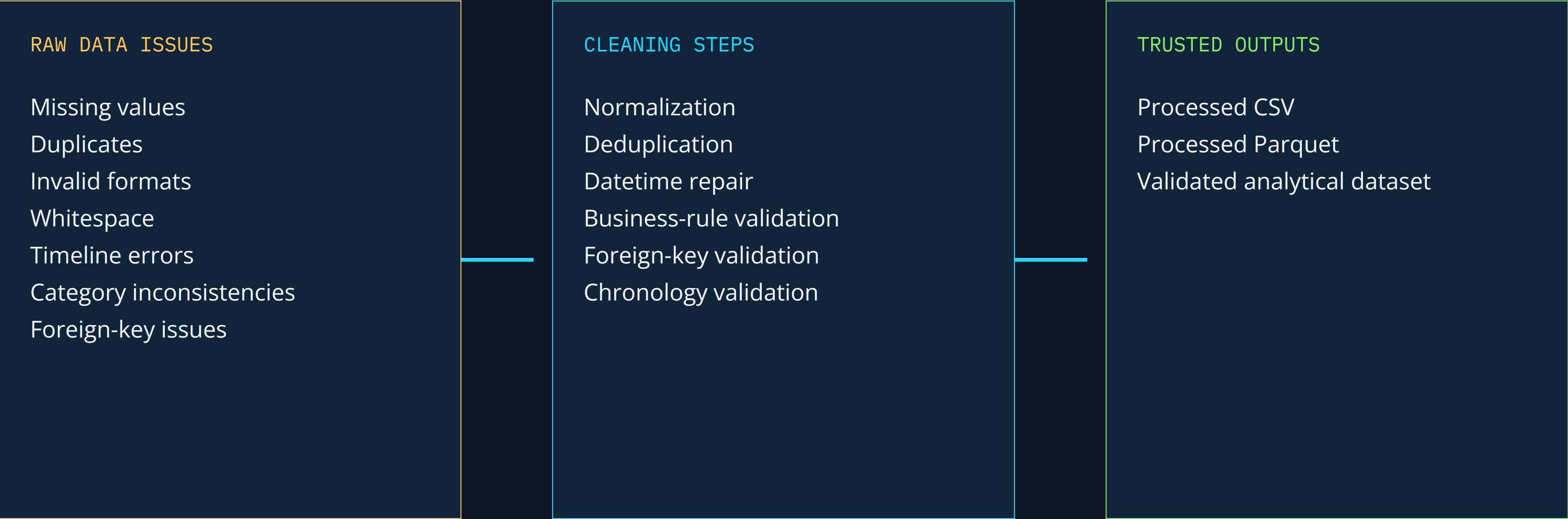
“Build an end-to-end analytical system that connects raw relational data with machine learning, interactive screening, and executive BI.”

# Synthetic Academic Integrity Dataset



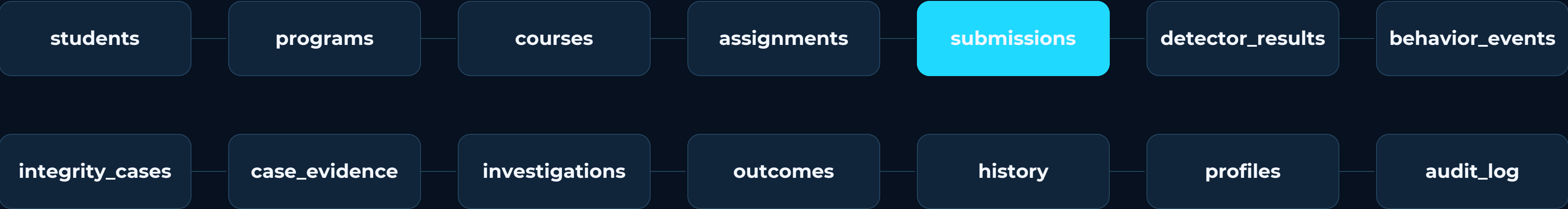
SYNTHETIC DATASET — CREATED FOR DEMONSTRATION

# From Messy Data to Trusted Data





# Relational Data Architecture



vw\_submission\_analytics • vw\_detector\_analysis • vw\_student\_integrity\_profile  
vw\_assignment\_analytics • vw\_integrity\_case\_analysis • vw\_behavior\_event\_analysis

14 NORMALIZED TABLES / PK + FK / INDEXES / CONSTRAINTS / AUDIT QUERIES

# AI-Generation Prediction Pipeline

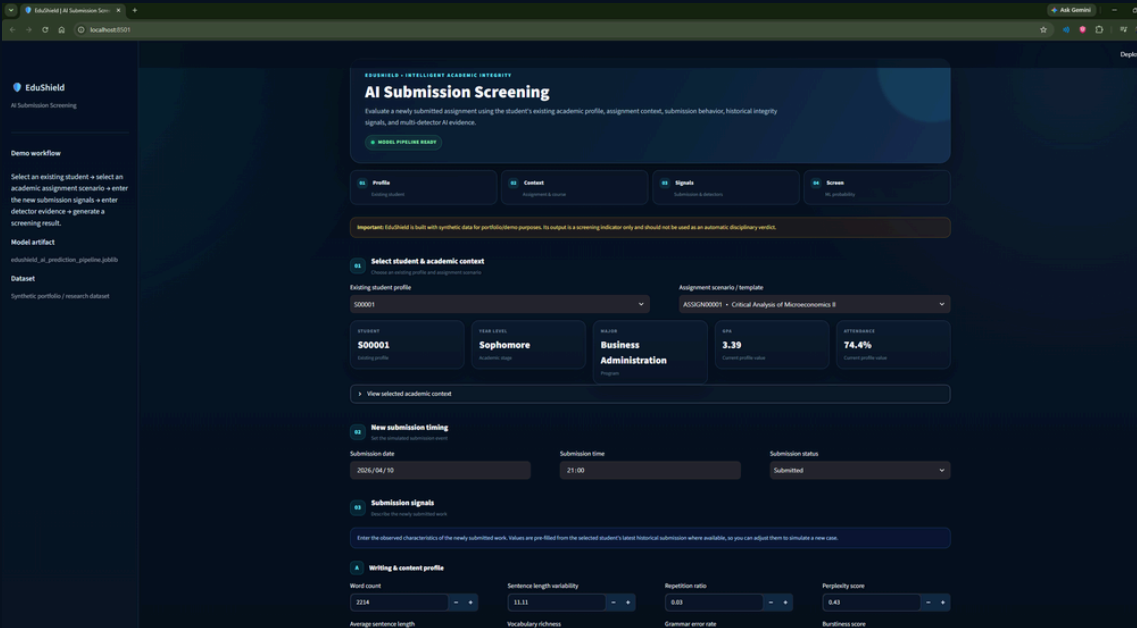


TARGET / is\_ai\_generated

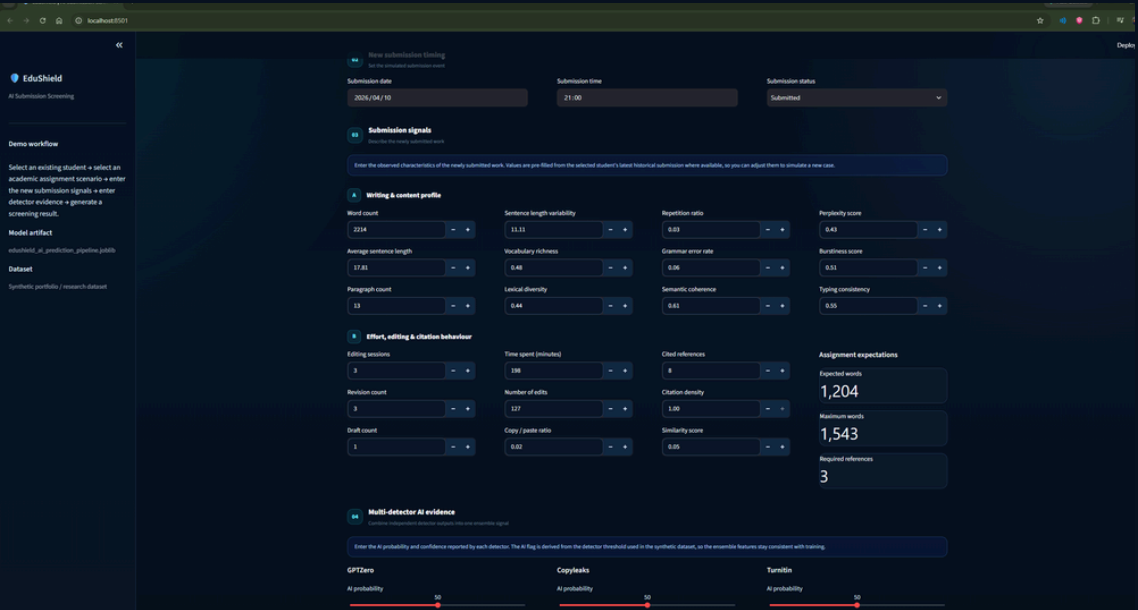
No accuracy or performance metrics shown.



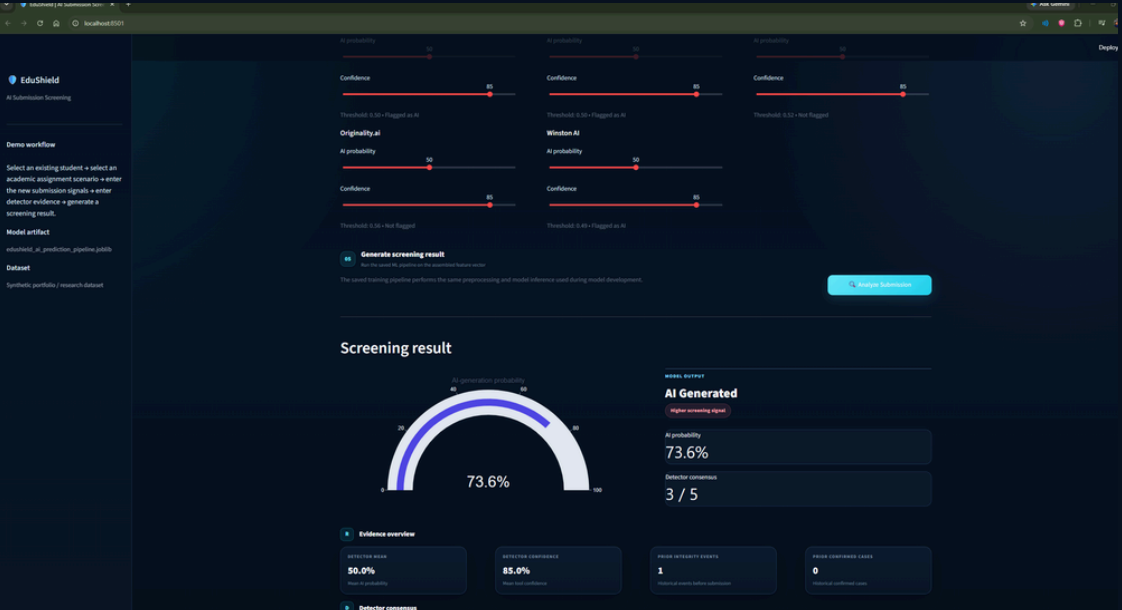
# Interactive AI Submission Screening



01 HOME PAGE



02 VALUES / INPUTS

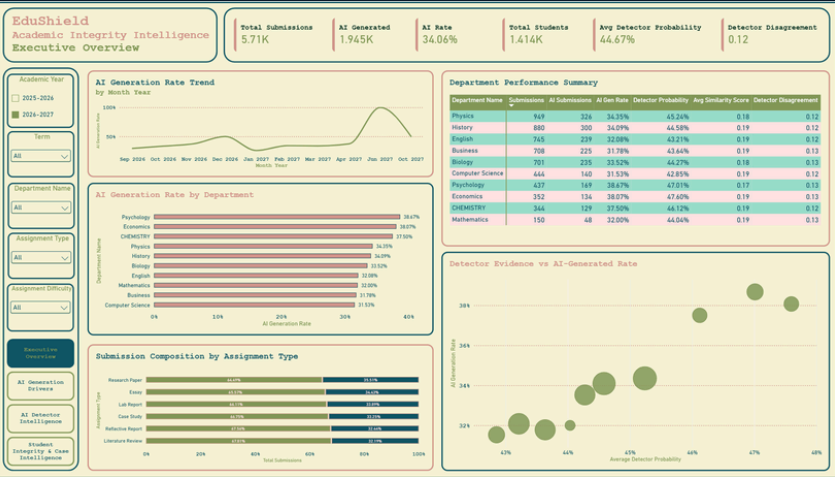


03 PREDICTION OUTPUT

Student Selection ↓ Assignment Context ↓ Submission Inputs ↓ Detector Evidence ↓ ML Prediction ↓ AI Probability + Supporting Signals

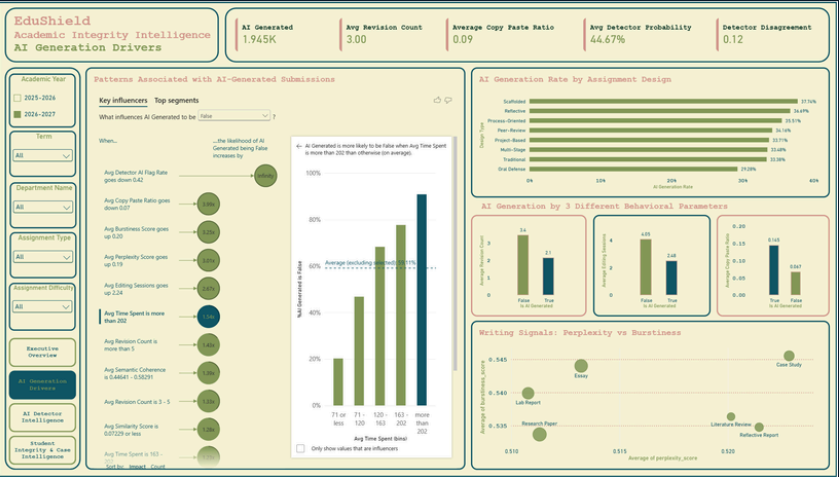


# Power BI Intelligence Layer



## 01 Executive Overview

High-level institutional submission and AI-generation insights



## 02 AI Generation Drivers

Explore features associated with AI-generated submissions



## 03 Detector Intelligence

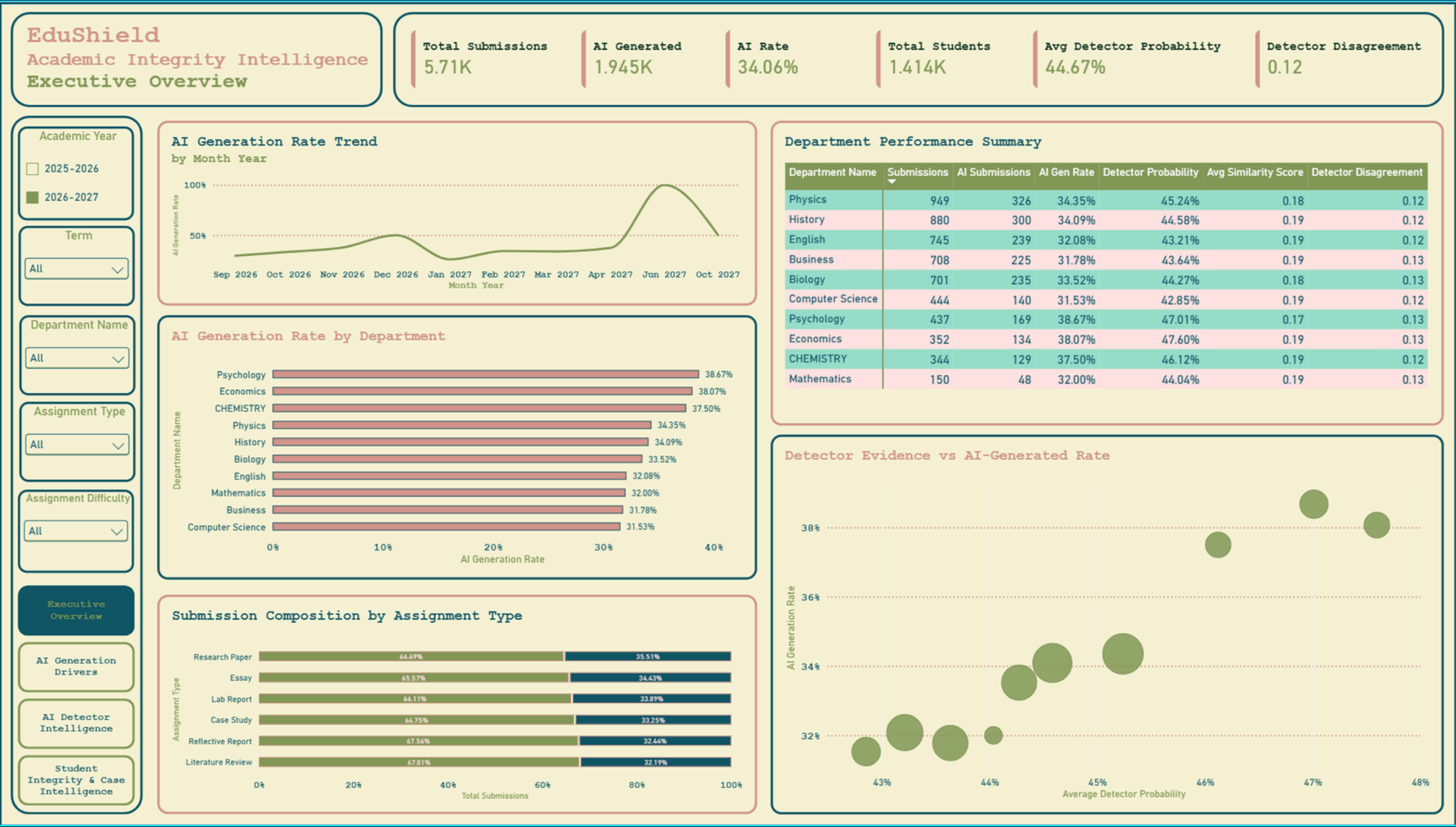
Compare five synthetic AI detectors' behavior and metrics



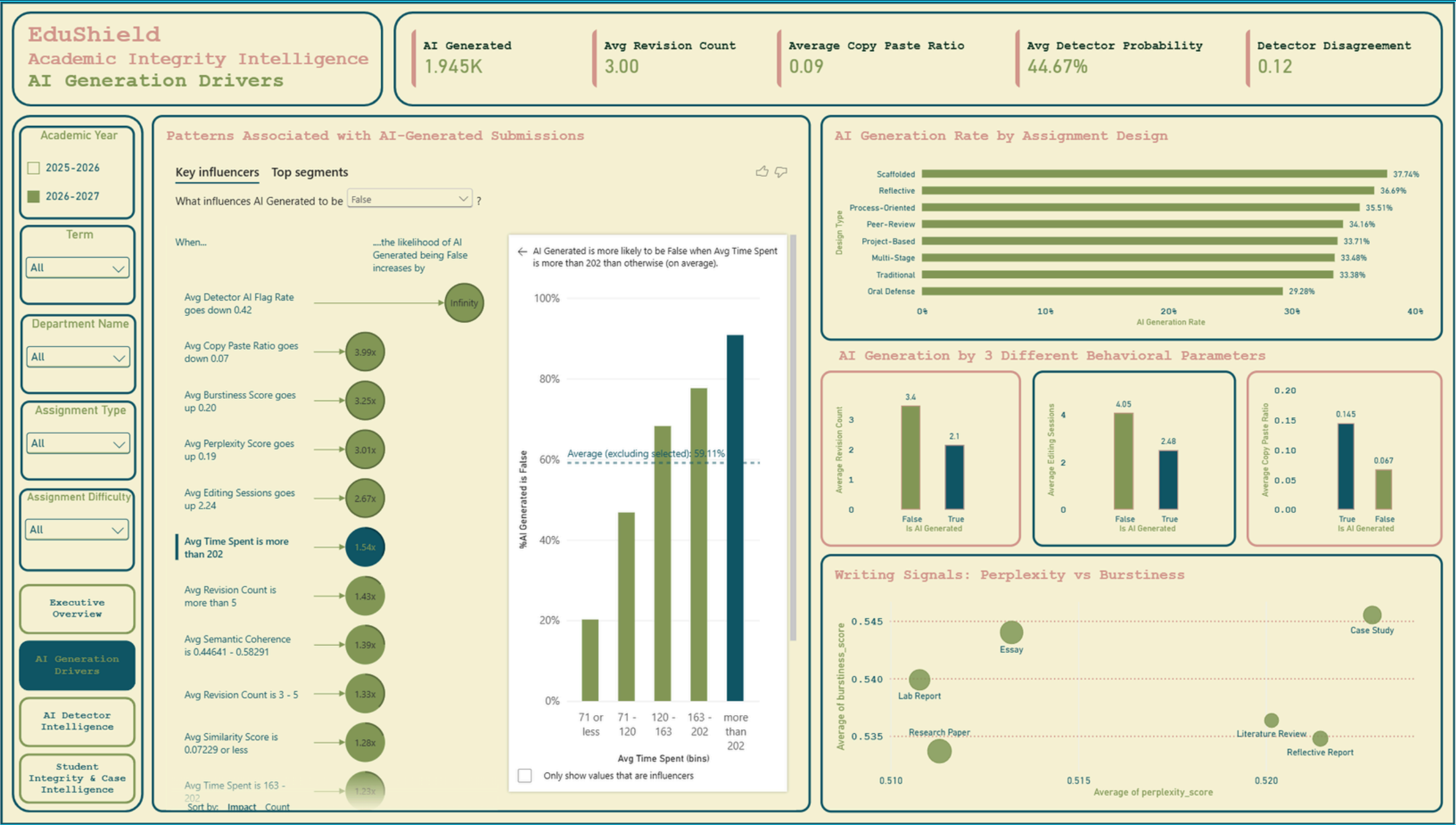
## 04 Student Integrity & Case Intelligence

Analyze student integrity profiles and case outcomes

# Power BI | Executive Overview

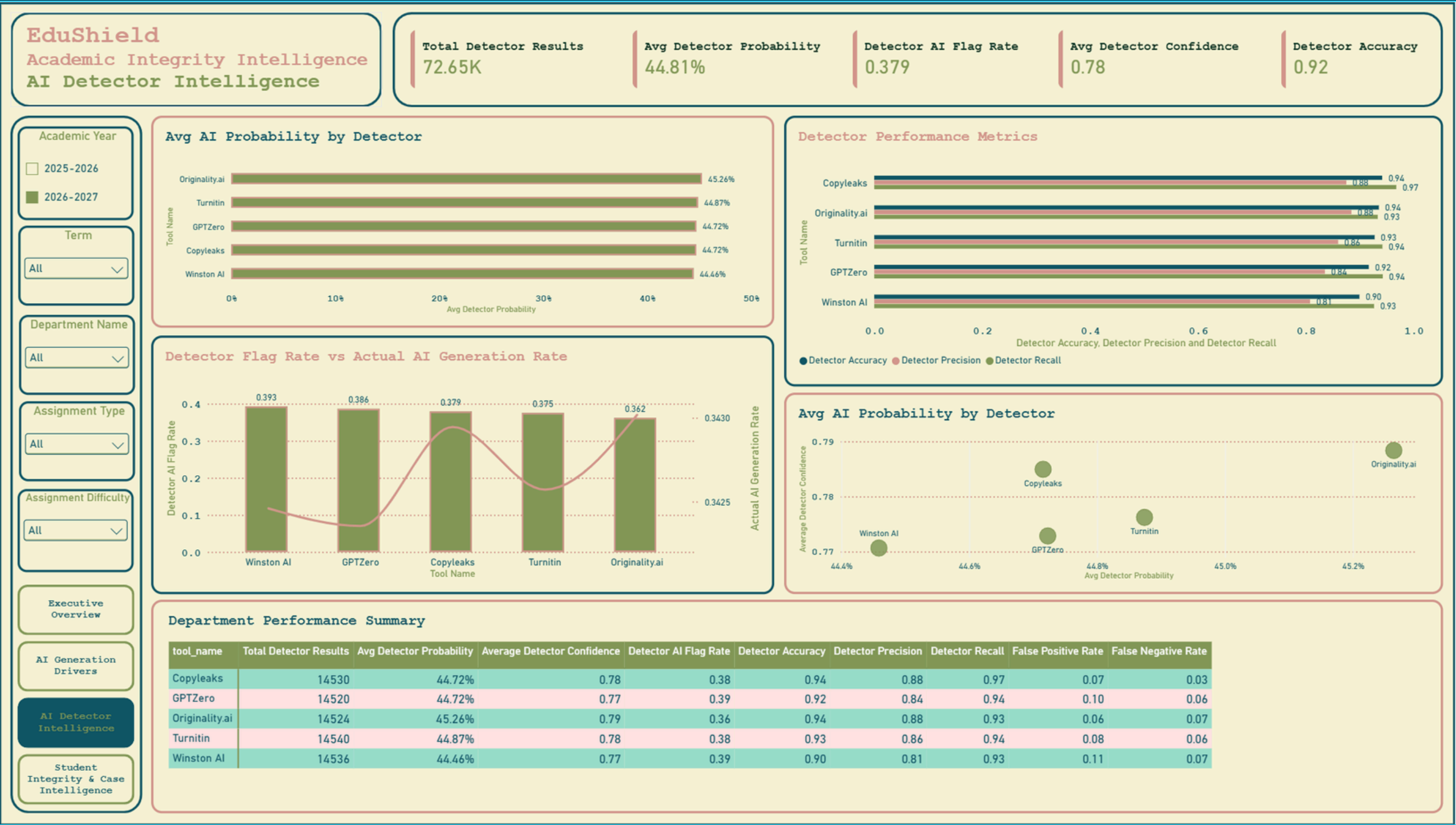


# Power BI | AI Generation Drivers



- Key Influencers
  - Assignment Characteristics
  - Revision Behavior
  - Copy/Paste Signals
  - Writing Signals
- Associations within synthetic data; not causal conclusions.

# Power BI | Detector Intelligence





# Power BI | Student Integrity & Case Intelligence



# Technology Stack

DATA

Python • Pandas • NumPy

DATABASE

MySQL • SQLAlchemy • PyMySQL

MACHINE LEARNING

scikit-learn • XGBoost • joblib

APPLICATION

Streamlit

BUSINESS INTELLIGENCE

Power BI

DEVELOPMENT

Jupyter • Git • GitHub

# What This Project Demonstrates

- 01 End-to-end data pipeline design
- 02 Relational data modeling
- 03 Realistic data-quality engineering
- 04 Leakage-aware ML feature engineering
- 05 Interactive BI and ML applications

EduShield connects data engineering, machine learning, application development, and business intelligence into one analytical ecosystem.



# Responsible Use & Limitations

01 **Synthetic data only**

02 **Synthetic detector outputs**

03 **Prediction is probabilistic, not deterministic**

04 **Key Influencers show associations, not causation**

05 **Indicators should not be used as standalone evidence**

06 **Human review required for any real-world decisions**

# From Raw Data to Actionable Intelligence

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**“EduShield demonstrates how a complex academic integrity workflow can be transformed into a structured, explainable, and interactive analytics platform.”**

# Thank You

EduShield — Academic Integrity Intelligence Platform

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GitHub Repository

[\[https://github.com/Aditya-Ujjwal/EduShield--Academic-Integrity-Intelligence-Platform.git\]](https://github.com/Aditya-Ujjwal/EduShield--Academic-Integrity-Intelligence-Platform.git)

Python • SQL • MySQL • Machine Learning • Streamlit • Power BI

ADITYA UJJWAL